

GNIOT Institute of Management Studies – PGDM (Batch 2024-26)
(TRIMESTER –I) END TERM EXAMINATION, OCTOBER-2024

Statistics For Managers

[Time: 2 Hours]

[Maximum Marks: 60]

[**BL** – Bloom's Taxonomy Levels (1- Remembering, 2- Understanding, 3 – Applying, 4 – Analysing, 5 – Evaluating, 6 - Creating)

CO – Course Outcomes]

Following are the course outcomes of the subject: - **Statistics For Managers**

CO	Course Outcome (CO)
CO1	Understand the fundamentals of statistics and its interdisciplinary role in decision making
CO2	Present and understand the data
CO3	Perform fundamental univariate and bivariate data analysis to draw meaningful insights

(Note- Faculty can add the more CO's as per their subject.)

SECTION-A

Attempt all parts of the following: (Short Answer Type Questions, do not exceed 200 words)

Q.1	Marks	CO	BL
(A)	4	CO – 1	L – 1
(B)	4	CO – 1	L - 1
(C)	4	CO – 1	L – 1
(D)	4	CO – 1	L – 3
(E)	4	CO – 1	L - 3

SECTION-B

Attempt all questions out of the following: (Long Answer Type Questions)

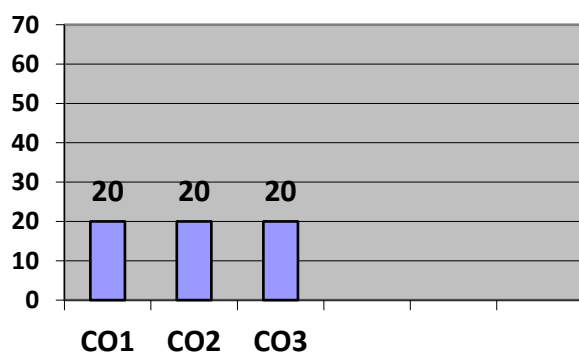
Q. No	Marks	CO	BL
2 (A)	10	CO – 2	L – 3
2 (B)	10	CO – 2	L – 3
3 (A)	10	CO – 2	L – 3
3 (B)	10	CO – 2	L – 3

SECTION-C

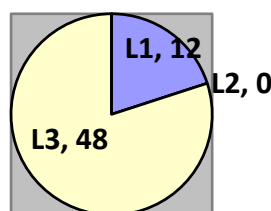
Attempt all parts of the following: (Caselet / Numerical)

Q.4	Marks	CO	BL
4 (A)	10	CO – 3	L – 3
4 (B)	10	CO – 3	L - 3

■ Course Outcome Wise Marks Distribution



■ L1 ■ L2 ■ L3



Bloom's Level wise Marks

Moderated By

Head- IQAC

Dean - Examinations

Printed Pages: 1

Roll No:

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Subject Code: **DM-103**

Paper ID: **11003**

GNIOT Institute of Management Studies – PGDM (Batch 2024-26)
(TRIMESTER – I) END TERM EXAMINATION, DECEMBER - 2024
Statistics for Managers

[Time: 2 Hours]

[Maximum Marks: 60]

INSTRUCTIONS:

- (i) There are **THREE** sections in this question paper.
- (ii) Attempt questions from each section as per the directions.
- (iii) Make suitable assumptions wherever necessary.

SECTION-A

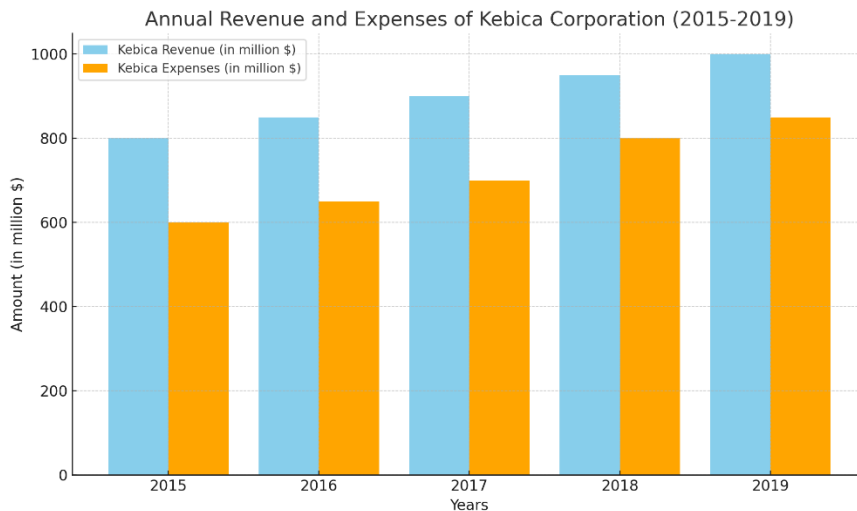
(Short Answer Type Questions, do not exceed 200 words)

Q.1. Attempt all parts of the following:

(5×4=20)

- (a) Consider a scenario where a company wants to understand its sales trends over the past year. What statistical tools and techniques would be most useful in this analysis, and why would these be effective in a business context?
- (b) Imagine you are tasked with presenting complex sales data to senior management. How would you categorize and visually present this data to ensure clarity and facilitate decision-making?
- (c) Envision a business scenario where understanding customer satisfaction is crucial. How would different measures of central tendency assist in summarizing customer feedback data, and what business decisions could be influenced by this analysis?
- (d) Design a student transportation survey consisting of 4 questions. The survey should include questions that demonstrate the following levels of measurement:
 - Two question using the nominal level of measurement
 - One question using the ordinal level of measurement
 - One question using the metric level of measurement.

(E) Analysis the Bar Graph given below and answer the question.



Q: Approximately what percentage of the total revenue over these five years is accounted for by the total expenses of Kebica Corporation? Based on this analysis, should Kebica Corporation focus more on increasing its revenue, reducing its expenses, or maintaining its current strategy to enhance profitability?.

SECTION-B

(Long Answer Type Questions)

Q.2. Attempt all questions of the following

(2×10=20)

- a) Following is the closing value of the NIFTY during the months of March and April 2023

March: 5719, 5699, 5784, 5819, 5863, 5946, 5942, 5914, 5851, 5909, 5873, 5835, 5746, 5694, 5658, 5651, 5634, 5641, 5683

April: 5704, 5748, 5673, 5575, 5553, 5543, 5495, 5598, 5594, 5529, 5568, 5689, 5689, 5783, 5734, 5837, 5916, 5872, 5904, 5930

Question: During which month was NIFTY Index more consistent? Looking at the NIFTY closing values for March 2023, can you find the value that separates the highest 25% of the data from the rest? What does this value reveal about the distribution of the NIFTY values for the month?

OR

- b) Calculate the skewness for the NIFTY closing values in both March and April. Based on your results, interpret the skewness and comment on the distribution shape for each month's data. Is the data positively skewed, negatively skewed, or approximately symmetric? If you were to divide the NIFTY closing values for March 2023 into 10 equal parts, which values would represent the top 10% and bottom 10% of the data? How do these values help you understand the extremes in the dataset?

Q.3. Attempt all questions of the following

The following table represents the monthly income (in ₹000s) of a sample of 50 individuals, grouped into income ranges:

Income Range (₹000s)	Number of Individuals
10 - 20	8
20 - 30	12
30 - 40	18
40 - 50	22
50 - 60	10
60 - 70	5
70 - 80	3

Q: To better structure the salary brackets, the HR team wants to find the midpoint in the salary distribution. What is the middle income range that divides the employees into two equal halves? The company's financial officer is curious about the average income in the organization to help benchmark salaries. What is this average income? Using the income data of 50 individuals, can you determine the income level below which 75% of the individuals fall? What does this value suggest about the distribution of income among the sample?

OR

The table below shows the number of hours employees at a company spend on skill development programs in a month:

Hours Range	Number of Employees
5 - 10	10
10 - 15	12
15 - 20	22
20 - 25	18
25 - 30	15
30-35	12
35-40	10

Q: The Company wants to find the point that divides the employees' training time into two equal parts. What range of hour's best represents this middle training time? HR noticed that the majority of employees tend to spend their time in a particular range of training hours. What is the most common range of hours employees dedicate to skill development each month? Based on the income data of the 50 individuals, can you identify the income level below which 90% of the individuals fall? How does this value help you understand the distribution of income in the sample?

SECTION-C

(Case study / Numerical Question)

A restaurant manager is investigating factors that might influence the monthly sales of their restaurant. The manager believes that two specific factors, the amount spent on social media advertising and the number of daily visitors, could have an impact on monthly sales but is unsure about the best statistical approach.

The dataset below provides monthly sales (in dollars) for the restaurant along with two potential influencing factors:

Month	Social Media Advertising Expenditure (\$)	Number of Daily Visitors	Monthly Sales (\$)
Jan	1500	320	24000
Feb	1600	340	25500
Mar	1700	360	27000

Month	Social Media Advertising Expenditure (\$)	Number of Daily Visitors	Monthly Sales (\$)
Apr	1400	300	22500
May	1800	370	28500
Jun	1600	340	26000
Jul	1500	330	24500
Aug	1300	310	22000
Sep	1900	380	29000
Oct	1800	370	27500
Nov	1600	350	25500
Dec	1500	340	25000

Q.4. Attempt all parts of the following:

(2×10=20)

4(a) Determine if there is any **relationship** between the advertising expenditure and monthly sales? Assess how the number of daily visitors **influences** monthly sales?

4(b) following table gives you the regression output

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.990155
R Square	0.980407
Adjusted R Square	0.976053
Standard Error	337.8028
Observations	12

ANOVA

	df	SS	MS	F	Significance F
Regression	2	51389670	25694835	225.1745	2.06276E-08
Residual	9	1026997	114110.7		
Total	11	52416667			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	1647.514	1970.302	0.836173	0.424701	-2809.618799	6104.647	-2809.618799	6104.647
Social Media Advertising Expenditure (\$)	7.47555	1.951497	3.830674	0.004024	3.06095658	11.89014	3.06095658	11.89014
Number of Daily Visitors	34.96333	13.77387	2.53838	0.031795	3.804659299	66.12199	3.804659299	66.12199

Q: Write the regression equation. Can we conclude that the Social media advertisement and Number of daily visitors have a significant relationship with the monthly sales?

Considering the magnitude of the coefficients and the p-values for both predictors (Social Media Advertising and Number of Daily Visitors), which factor appears to have a greater influence on monthly sales? How do you come to this conclusion?